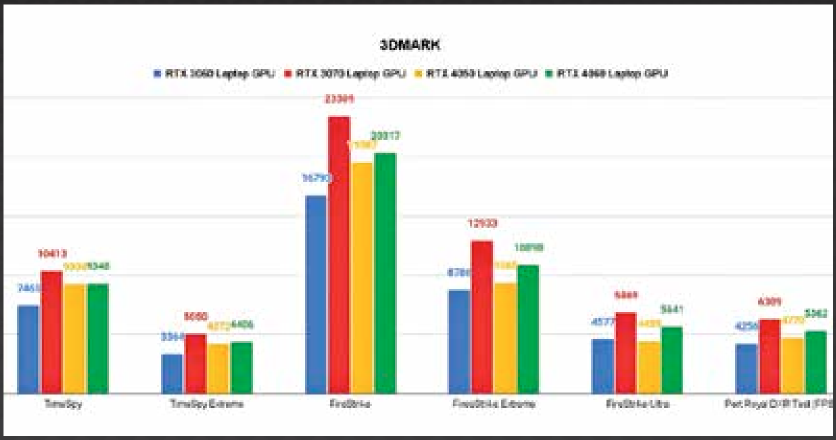
featuring higher tier GPUs like an RTX 4050, similarly for the RTX 3070 and the RTX 4060.

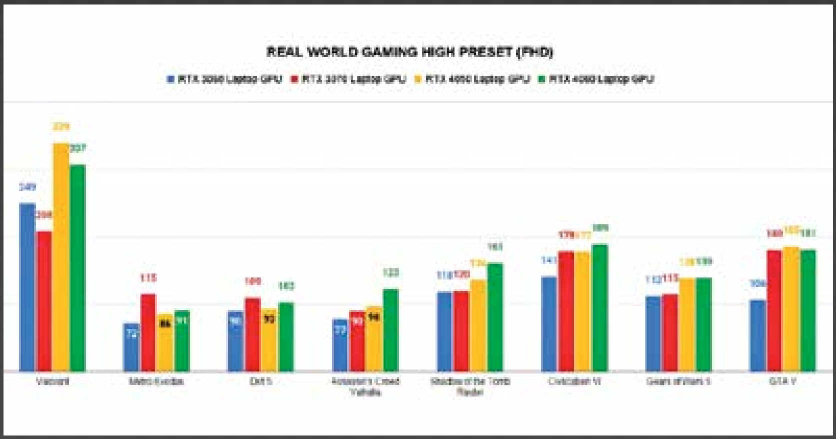
The idea behind this article is to help you decide on which laptop GPU you should go for, that will offer both better performance and future-proofing for the next few years. For this comparison, we'll be comparing laptop GPUs which feature the highest TGP allowed by NVIDIA.

With that out of the way, let's take a look at our contenders. For all intents and purposes, these are the best laptops we've tested so far featuring these GPUs.

The RTX 3070 features the highest TGP among the cards since the xx70 series cards technically fall under "high-end" GPUs. So it does have a leg up over every other card here in terms of sheer power.

Another thing to note before we get into this comparison is that all of these laptops have different processors. While they're not vastly different in terms of performance, you will notice some differences in FPS numbers when it comes to games that are processor-intensive. It shouldn't be by much; the GPUs will still have the largest impact on performance numbers here. With that said, let's get started!





Synthetic GPU performance

We start things off with the synthetic GPU benchmark. We use the 3DMark benchmark to test DirectX 11, DirectX12, and Ray Tracing performance.

As you can see from the graph, the RTX 3070 outperforms every GPU in this comparison. If you consider its TGP of 140W, it makes sense, and there is some merit to it being labelled a higher-end GPU. Everything else is about as expected. You can see the newer RTX 4050 gives better DirectX12 performance than the older RTX 3060.

However, the RTX 3060 does better than the RTX 4050 in DirectX11 performance. This is relevant as newer games will run on DX12, but you would get better performance on the RTX 3060 from older titles that do not have or support DirectX 12. The RTX 4060 offers similar DX12 performance to the RTX 4050, but does significantly better in DX11 performance with only the RTX 3070 outperforming it.

Real-world gaming performance

With that, we come to real-world gaming performance. While synthetic benchmarks are a decent enough way to gauge a laptop's gaming performance, real-world gaming performance is ultimately what matters. All of our real-world gaming tests are done in FHD (1920x1080p) resolutions, and we use the game's preset high and medium settings.

Based on our synthetic GPU observations, the RTX 3070 should be bagging the win here, but real-world gaming performance numbers tell a different story.

We start things off with the high preset of real-world gaming benchmarks. Despite being the highest-scoring GPU in synthetic benchmarks, the RTX 3070 scores the highest FPS numbers in only two titles. The RTX 4060 comes out as the best performer here with the RTX 4050 offering some serious competition. In fact, the RTX 4050 easily outperforms the RTX 3060 in every single gaming benchmark. Note that all of these titles have DirectX12 support. We should also point out that the laptop with the RTX 4050 in this comparison features the most power-